

Adventure Math Game – Kivy Mobile Code

Complete Python/Kivy source code for the mobile-compatible game.

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import random

from kivy.app import App
from kivy.ui.boxlayout import BoxLayout
from kivy.ui.label import Label
from kivy.ui.button import Button
from kivy.ui.textinput import TextInput
from kivy.clock import Clock

class MathGame(BoxLayout):

    def __init__(self, **kwargs):
        super().__init__(
            orientation="vertical",
            padding=25,
            spacing=15,
            **kwargs
        )

        self.score = 0
        self.question_no = 0
        self.total_questions = 10
        self.time_left = 20
        self.answer = 0
        self.timer = None
        self.player = ""

        self.start_screen()

    # START SCREEN
    def start_screen(self):

        self.clear_widgets()

        self.add_widget(Label(
            text="■ ADVENTURE MATH GAME",
            font_size="28sp",
            bold=True
        ))

        self.add_widget(Label(
            text="Learn Mathematics by Playing!",
            font_size="18sp"
        ))

        self.name = TextInput(
            hint_text="Enter your name",
            multiline=False,
            font_size="20sp",
            size_hint_y=None,
            height=60
        )

        self.add_widget(self.name)

        button = Button(
            text="START GAME",
            font_size="22sp",
            bold=True,
            size_hint_y=None,
            height=70
        )

        button.bind(on_press=self.start_game)

        self.add_widget(button)

    # START GAME
    def start_game(self, instance):

        self.player = self.name.text.strip()

        if not self.player:
            self.message.text = "Please enter your name."
            return

        self.score = 0
        self.question_no = 0

        self.game_screen()
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        self.next_question()

# GAME SCREEN
def game_screen(self):

    self.clear_widgets()

    self.info = Label(
        text="",
        font_size="18sp",
        bold=True
    )

    self.add_widget(self.info)

    self.timer_label = Label(
        text="■■ Time: 20",
        font_size="22sp",
        bold=True
    )

    self.add_widget(self.timer_label)

    self.question = Label(
        text="",
        font_size="30sp",
        bold=True
    )

    self.add_widget(self.question)

    self.answer_box = TextInput(
        hint_text="Enter answer",
        multiline=False,
        input_filter="int",
        font_size="24sp",
        halign="center",
        size_hint_y=None,
        height=60
    )

    self.add_widget(self.answer_box)

    self.submit = Button(
        text="SUBMIT ANSWER",
        font_size="20sp",
        bold=True,
        size_hint_y=None,
        height=65
    )

    self.submit.bind(
        on_press=self.check_answer
    )

    self.add_widget(self.submit)

    self.message = Label(
        text="",
        font_size="18sp",
        bold=True
    )

    self.add_widget(self.message)

# NEXT QUESTION
def next_question(self, *args):

    if self.question_no >= self.total_questions:
        self.end_game()
        return

    self.question_no += 1
    self.time_left = 20

    self.info.text = (
        f"Player: {self.player}\n"
        f"Question: {self.question_no}/"
        f"{self.total_questions} "
        f"Score: {self.score}"
    )

    self.answer_box.text = ""
    self.submit.disabled = False
    self.message.text = ""

    self.make_question()

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        self.start_timer()

# QUESTIONS
def make_question(self):

    # EASY
    if self.question_no <= 3:

        a = random.randint(1, 50)
        b = random.randint(1, 50)

        op = random.choice(["+", "-"])

        if op == "+":
            self.answer = a + b
        else:
            self.answer = a - b

        self.question.text = f"{a} {op} {b} = ?"

    # MEDIUM
    elif self.question_no <= 7:

        a = random.randint(2, 20)
        b = random.randint(2, 10)

        op = random.choice(["x", "÷"])

        if op == "x":

            self.answer = a * b

            self.question.text = (
                f"{a} × {b} = ?"
            )

        else:

            self.answer = a

            self.question.text = (
                f"{a * b} ÷ {b} = ?"
            )

    # HARD
    else:

        x = random.randint(1, 20)
        b = random.randint(1, 20)

        self.answer = x

        self.question.text = (
            f"x + {b} = {x + b}\n\n"
            "Find x"
        )

# TIMER
def start_timer(self):

    if self.timer:
        self.timer.cancel()

    self.time_left = 20

    self.timer = Clock.schedule_interval(
        self.update_timer,
        1
    )

def update_timer(self, dt):

    self.timer_label.text = (
        f"■ Time: {self.time_left}"
    )

    self.time_left -= 1

    if self.time_left < 0:

        self.timer.cancel()

        self.submit.disabled = True

        self.message.text = (
            f"■ Time's Up!\n"
            f"Answer: {self.answer}"

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        )

        Clock.schedule_once(
            self.next_question,
            1.5
        )

# CHECK ANSWER
def check_answer(self, instance):

    if self.time_left < 0:
        return

    text = self.answer_box.text.strip()

    if not text:
        self.message.text = "Enter an answer!"
        return

    try:
        user_answer = int(text)
    except:
        self.message.text = "Enter a number!"
        return

    if user_answer == self.answer:

        self.score += 10

        self.message.text = (
            "■ Correct! +10 points"
        )

    else:

        self.message.text = (
            f"■ Wrong!\n"
            f"Correct answer: {self.answer}"
        )

    self.submit.disabled = True

    if self.timer:
        self.timer.cancel()

    Clock.schedule_once(
        self.next_question,
        1.5
    )

# END GAME
def end_game(self):

    if self.timer:
        self.timer.cancel()

    self.clear_widgets()

    self.add_widget(Label(
        text="■ GAME COMPLETED!",
        font_size="30sp",
        bold=True
    ))

    self.add_widget(Label(
        text=(
            f"{self.player}\n\n"
            f"YOUR SCORE\n"
            f"{self.score} / 100"
        ),
        font_size="28sp",
        bold=True
    ))

    if self.score >= 80:
        result = "■ Excellent!"

    elif self.score >= 60:
        result = "■ Very Good!"

    elif self.score >= 40:
        result = "■ Good! Keep Practicing!"

    else:
        result = "■ Keep Learning!"

    self.add_widget(Label(

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        text=result,
        font_size="22sp"
    ))

    again = Button(
        text="■ PLAY AGAIN",
        font_size="20sp",
        size_hint_y=None,
        height=65
    )

    again.bind(
        on_press=lambda x: self.start_screen()
    )

    self.add_widget(again)

    exit_button = Button(
        text="EXIT",
        font_size="20sp",
        size_hint_y=None,
        height=65
    )

    exit_button.bind(
        on_press=lambda x: App.get_running_app().stop()
    )

    self.add_widget(exit_button)

class MathGameApp(App):

    def build(self):
        return MathGame()

if __name__ == "__main__":
    MathGameApp().run()

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